EdgeVisor

PROVISION | CONFIGURE | **UPDATE**

Unified Device Management
for Smarter Industrial Operations

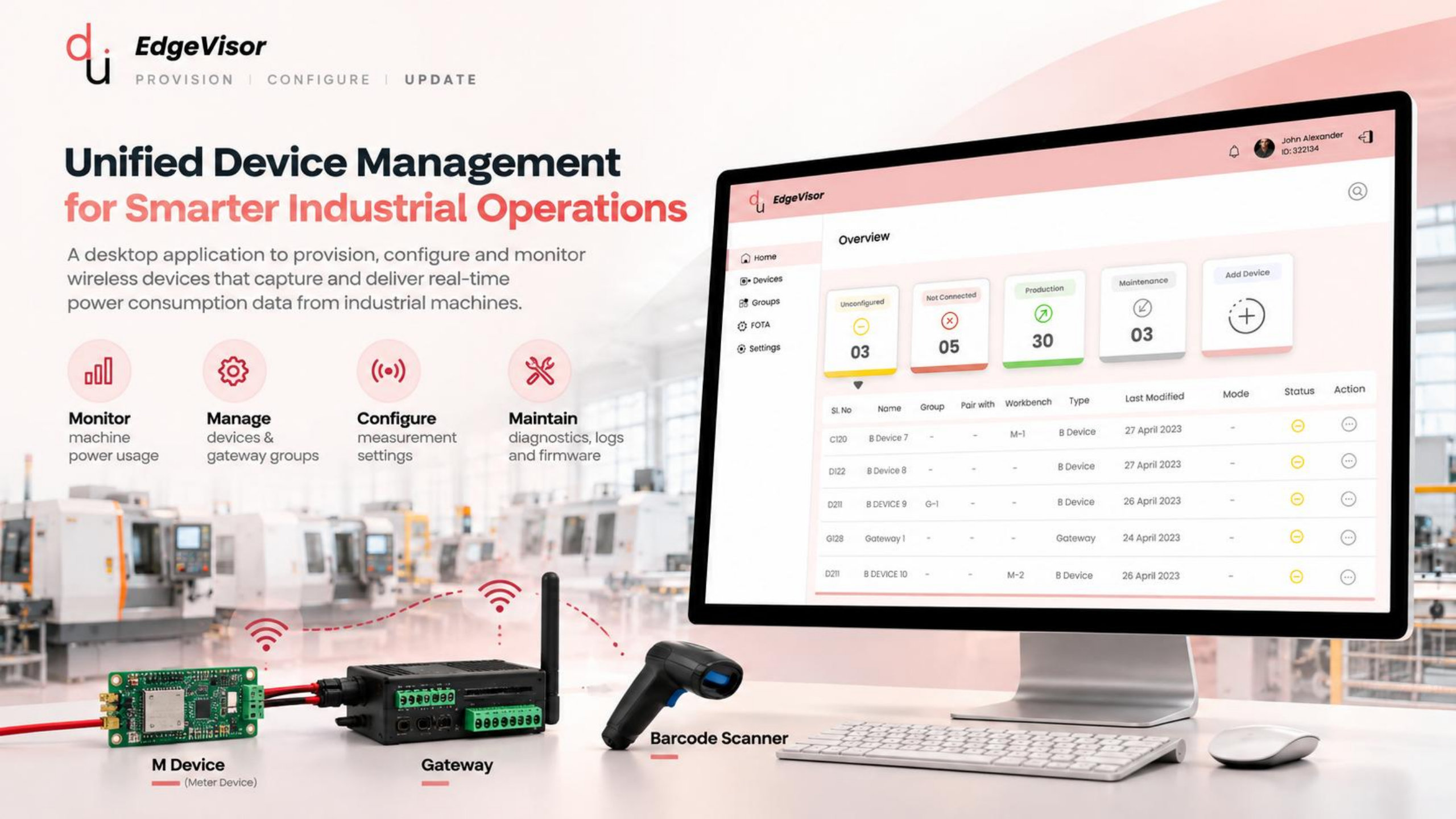
A desktop application to provision, configure and monitor wireless devices that capture and deliver real-time power consumption data from industrial machines.

Monitor machine power usage

Manage devices & gateway groups

Configure measurement settings

Maintain diagnostics, logs and firmware



A desktop workspace for connected-device operations

EdgeVisor helps factory personnel provision and manage power-monitoring devices from a central computer.

Client

Device Unlimited

Role

End-to-end product design

Status

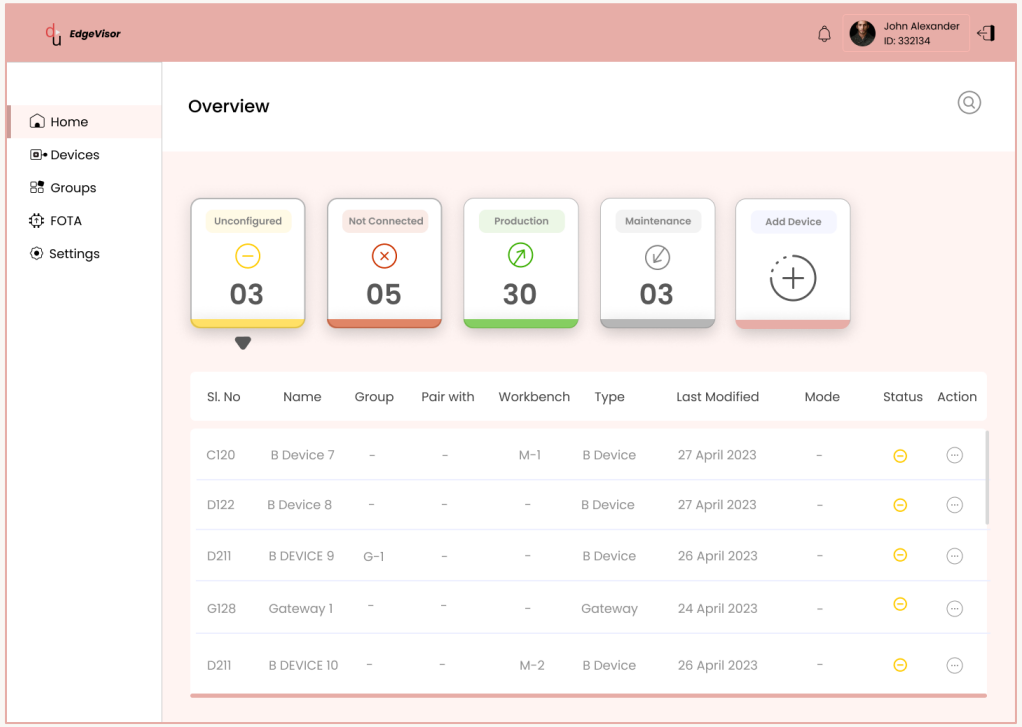
Developed and used

Inputs

Client and team requirements

The product in one sentence

EdgeVisor turns a connected hardware setup into a manageable software workflow: register boards, organize Gateway groups, pair M Devices, configure measurements, monitor states, and maintain devices.



Connect hardware realities to a usable product experience

My role spanned technical discovery, product decisions, and the end-to-end interface design.

Hardware collaboration

Learned how the meter boards, Gateway, M Device, Configurator, barcode scanner, and machine-side connections fit together.

Embedded + product teams

Worked across teams to understand device behavior, requirements, and the steps needed to bring the system into operation.

Product design

Translated the technical setup and operational needs into a desktop experience spanning onboarding, pairing, configuration, monitoring, and maintenance.

What this case study demonstrates

Systems thinking • Workflow design • Technical collaboration • Information architecture • UI design • Operational states and error-recovery considerations

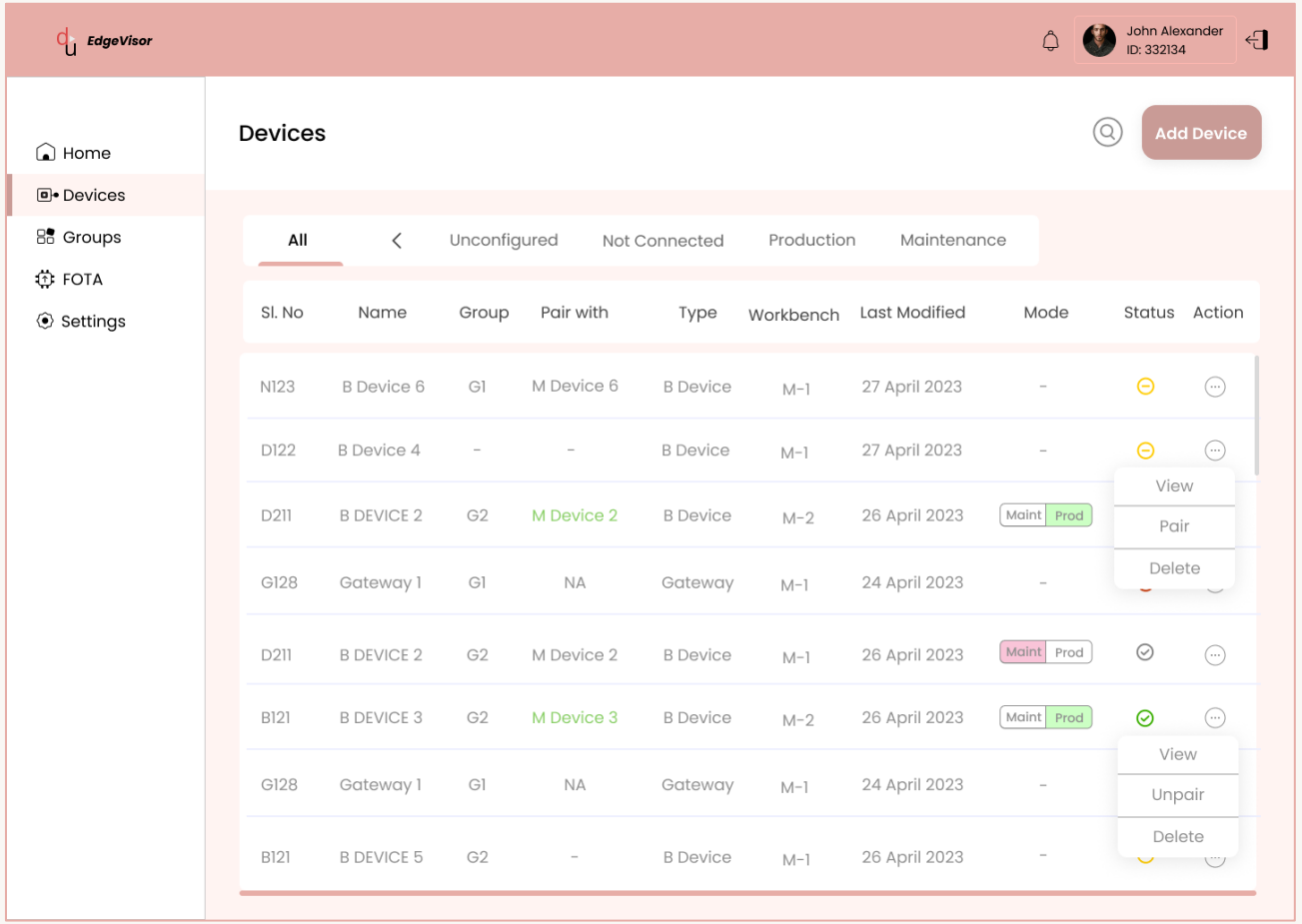
The challenge was bigger than a dashboard

Factory personnel need to set up and oversee distributed devices while the machines continue operating.

Design question

How might we make a multi-step hardware setup understandable and manageable from one computer, without requiring an operator to interact with every board for routine tasks?

- Many physical components and device relationships
- A sequence of setup steps before measurement starts
- Different operational states and maintenance actions
- Need to surface status and device details clearly



Start with system discovery and stakeholder requirements

The documented inputs are client/team requirements and cross-functional work with hardware and embedded teams.

1. Understand the system

Map the physical components, connections, and device roles.

2. Understand operations

Identify the tasks needed to register, pair, configure, monitor, and maintain devices.

3. Translate requirements

Turn technical dependencies into an ordered application workflow.

Design for the person overseeing machines from one place

PRIMARY USER

Factory operator / admin

Oversees machine power monitoring, device setup, and ongoing device health from a central workstation.

Needs to do

Register hardware, associate devices correctly, and verify readiness.

Needs to know

Which devices are unconfigured, disconnected, in production, or under maintenance.

Needs to recover

Diagnose connectivity, inspect logs, test data, and update firmware.

Environment

Industrial setting; multiple machines and device relationships; centralized oversight.

Define what a successful setup must enable

Translate the hardware and operational requirements into observable product tasks.

Provision

Register M Devices and Gateways by scanning QR codes.

Organize

Create a group and assign the Gateway.

Pair

Associate an M Device with the correct group and linked Gateway.

Configure

Set Gateway measurement state and M Device parameters.

Operate

Move configured devices into Production and start power consumption tracking.

Maintain

Run diagnostics, review logs, test captured data, and update firmware.

The operator moves from unregistered hardware to live measurement

User Goal

01

Prepare

Connect Configurator and scanner

02

Register

Scan M Device + Gateway QR codes

03

Group

Create Gateway group

04

Pair

Assign M Device to group

05

Configure

Set Gateway + meter parameters

06

Monitor

Verify Production state and power data

Potential friction points to address in the design

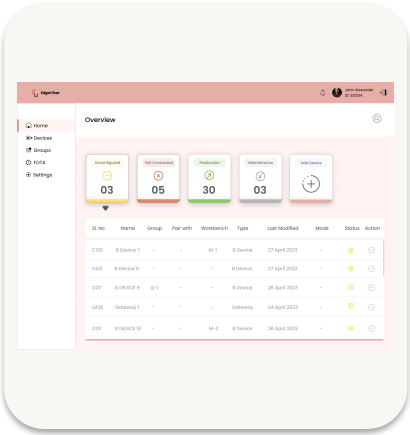
- Remembering dependencies between Gateway groups and M Devices
- Knowing whether a device is ready for the next step
- Recovering when a scan, connection, or configuration does not succeed

Organize the interface around lifecycle tasks and device states

The five navigation areas for successful operations

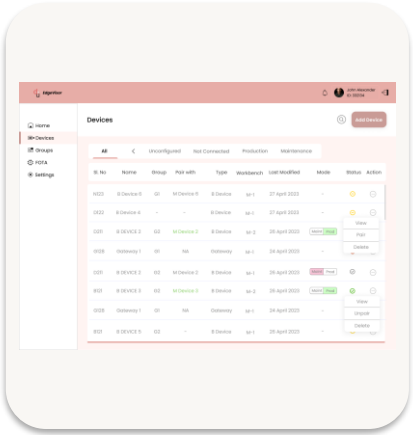
Home

Status overview



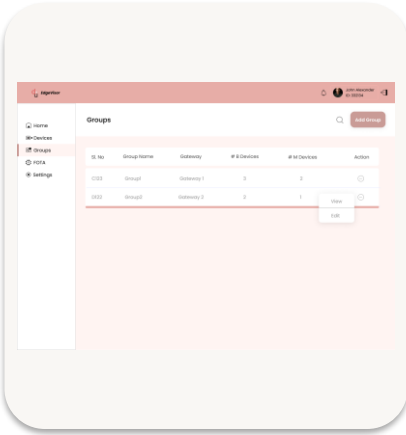
Devices

Register + configure



Groups

Gateway & M Device relationships



FOTA

Firmware updates



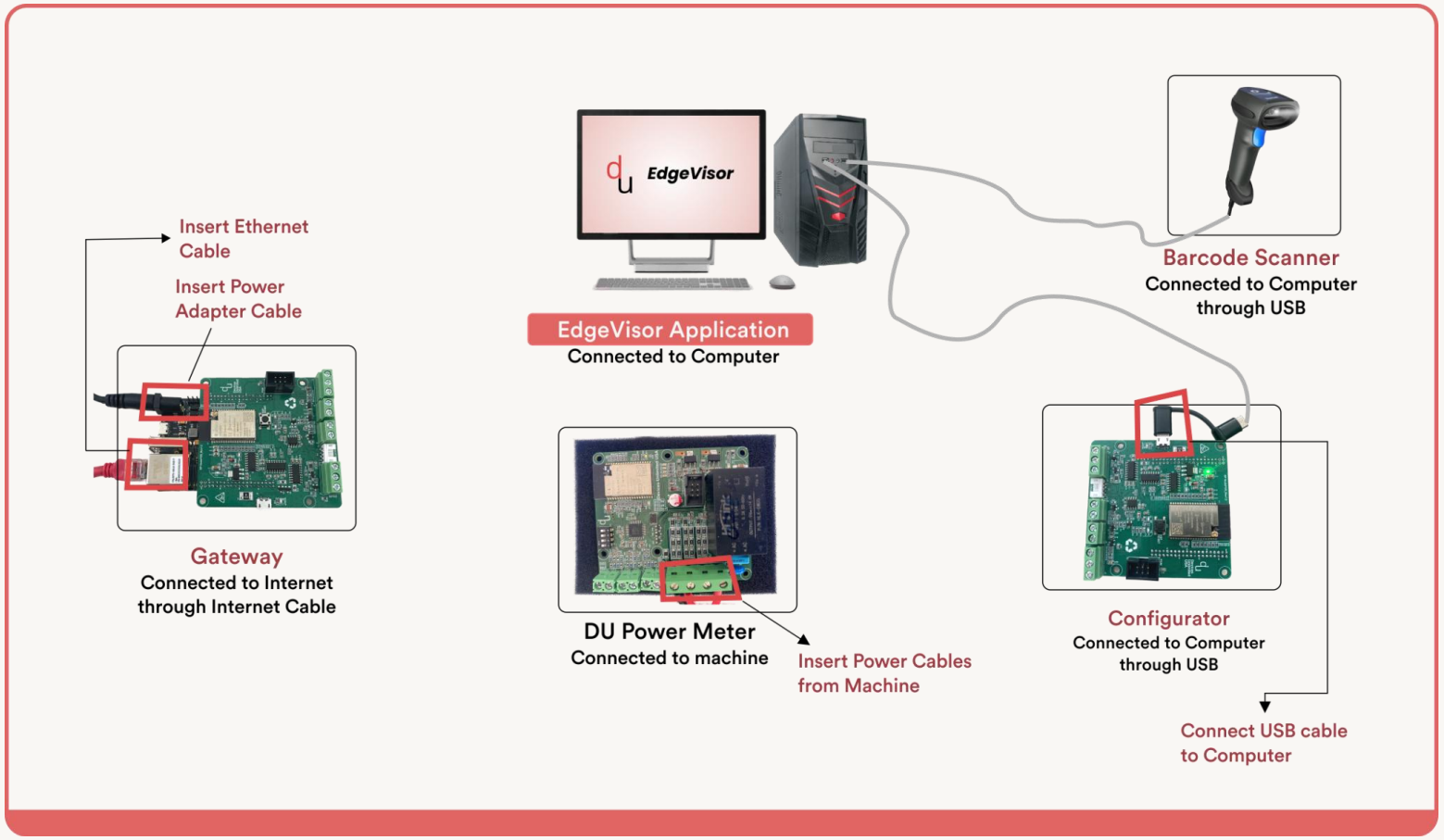
Settings

Application options



Understand the hardware before designing the workflow

The supplied connection diagram shows the computer, barcode scanner, Gateway, Configurator, and machine-side board.



Computer + EdgeVisor

Central interface for setup and monitoring.

Barcode scanner

Reads device QR codes during registration.

Configurator

Receives data from paired M Devices and Gateway. It connects to the computer.

M Device & Gateway

Board-level components involved in measurement and onboarding.

From the machine to the operator's workstation

Machine

Power consumption is supplied

→

M Device

Captures measurement data

→

Gateway

Receives data from paired M Devices

→

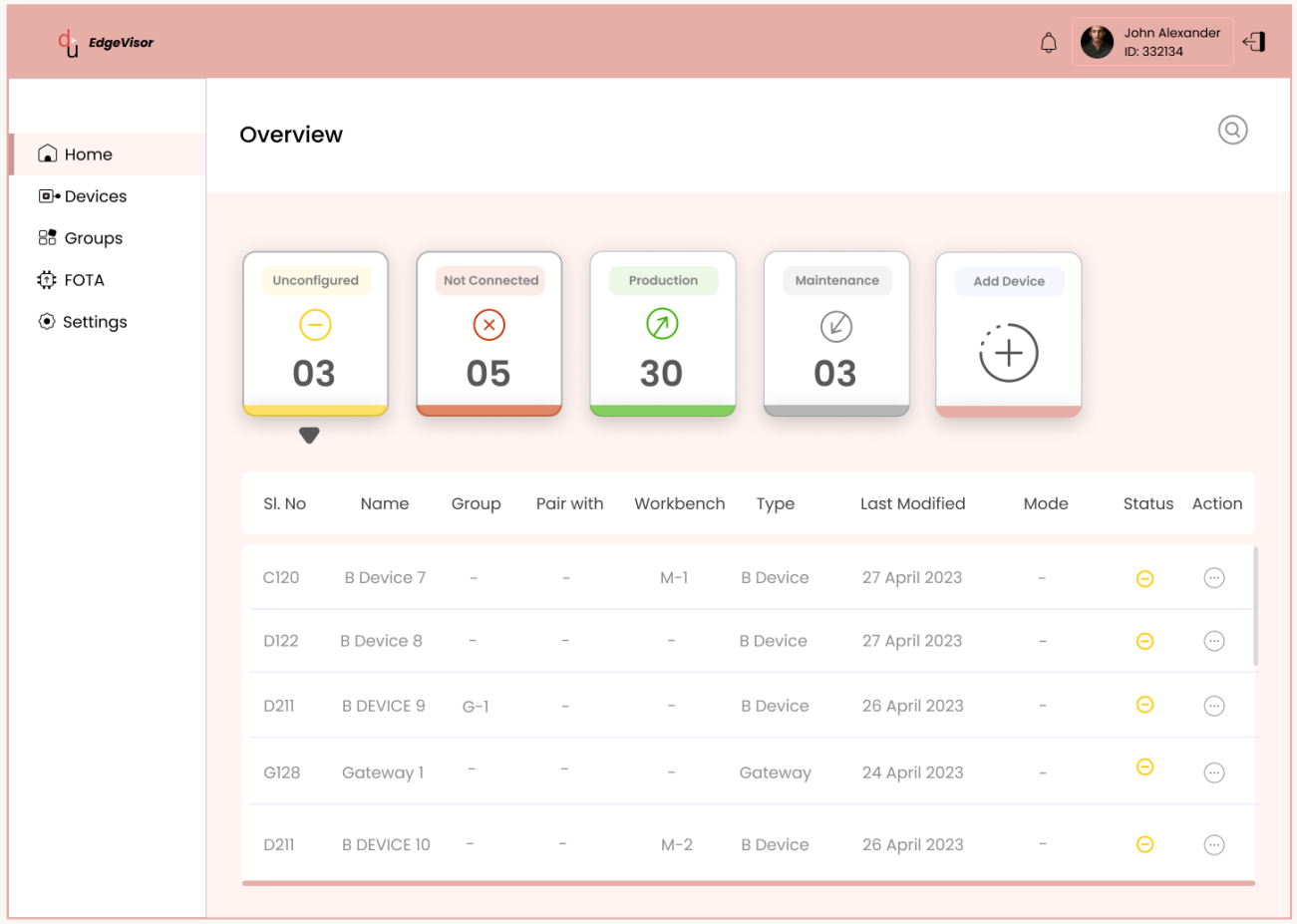
EdgeVisor

Operator configures and monitors devices through connected configurator

A Gateway group can include multiple M Devices. The application configures that relationship and provides a central place to review device states and access maintenance actions.

Home: surface state before the operator opens a device

Design overview: the dashboard combines summary cards, device-state navigation, and a detailed table.



At-a-glance status

Counts by state provide a quick overview.

Drill into records

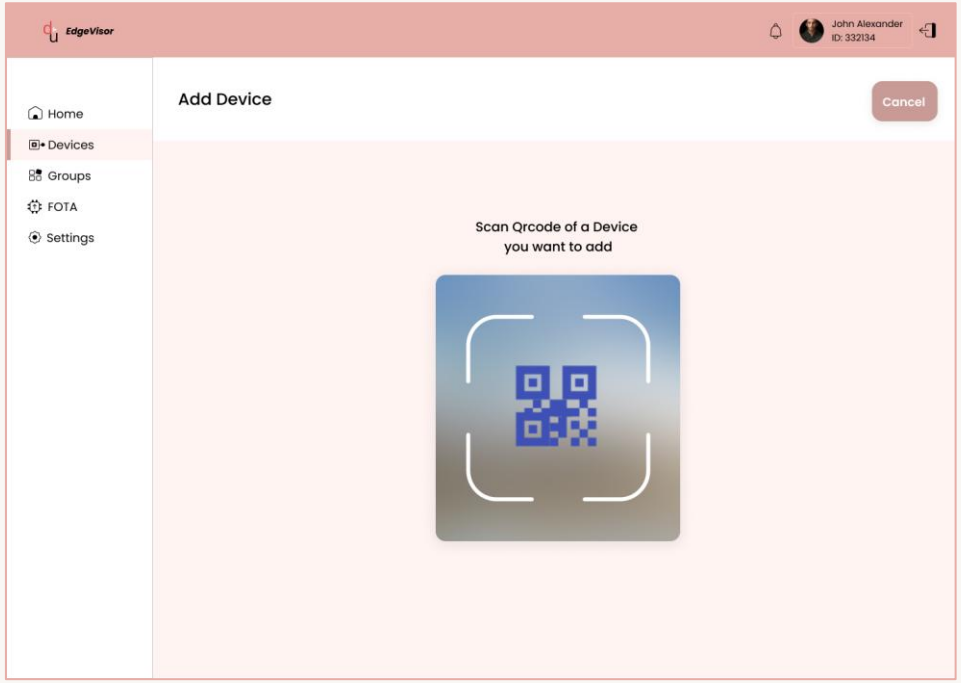
State tabs filter the device list.

Direct entry point

Add Device supports a new registration flow.

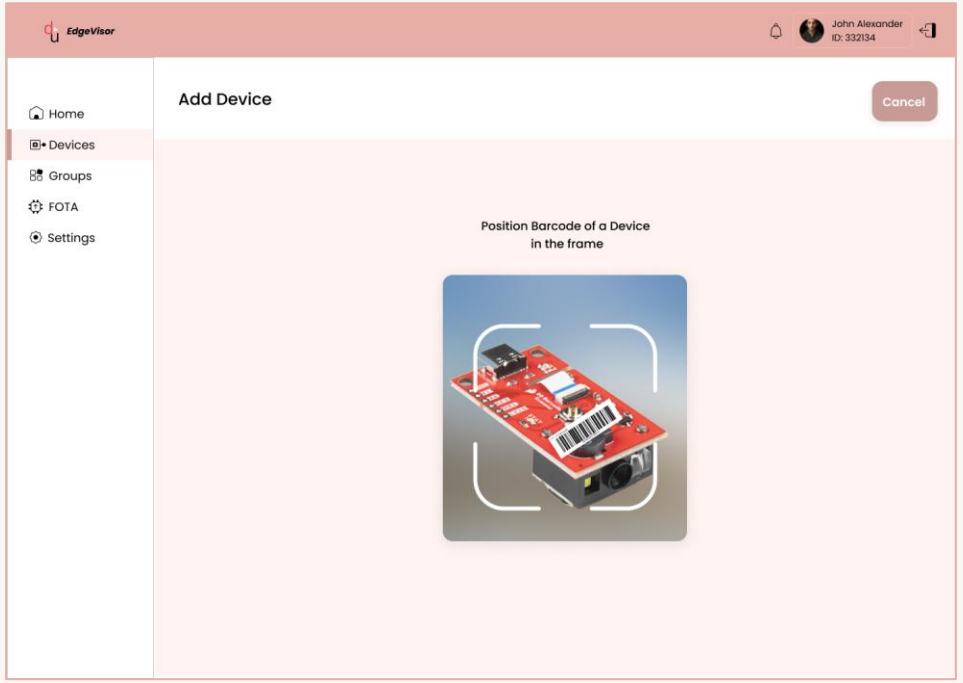
Register hardware with a scan-first flow

Make sure Configurator and barcode scanner must be connected and working.



01 Open Add Device

Start from Home or Devices. Confirm the Configurator is connected to the computer.

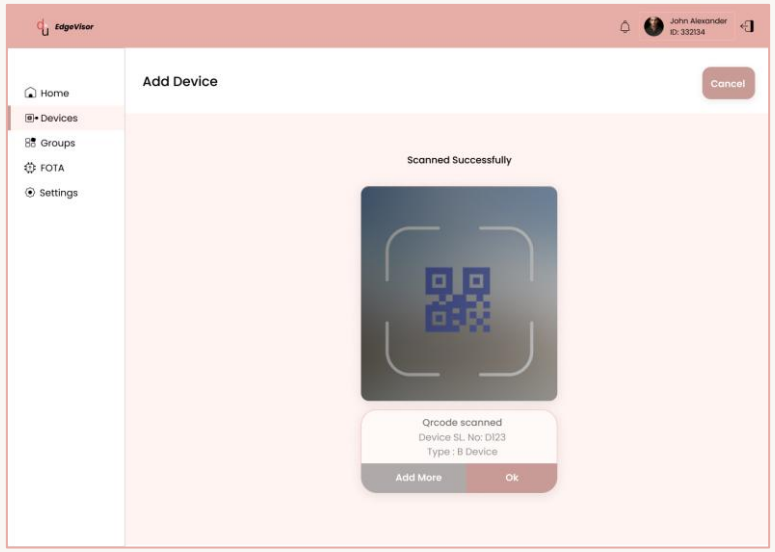


02 Scan QR code

Scan the QR code on a new M Device or Gateway using the connected scanner.

Confirm the scan and prepare both device types

The product needs at least one M Device and one Gateway before the setup can continue.



03 / Confirm

Check the success message after scanning.

04 / Add more

Use Add More to register additional devices.

05 / Verify setup

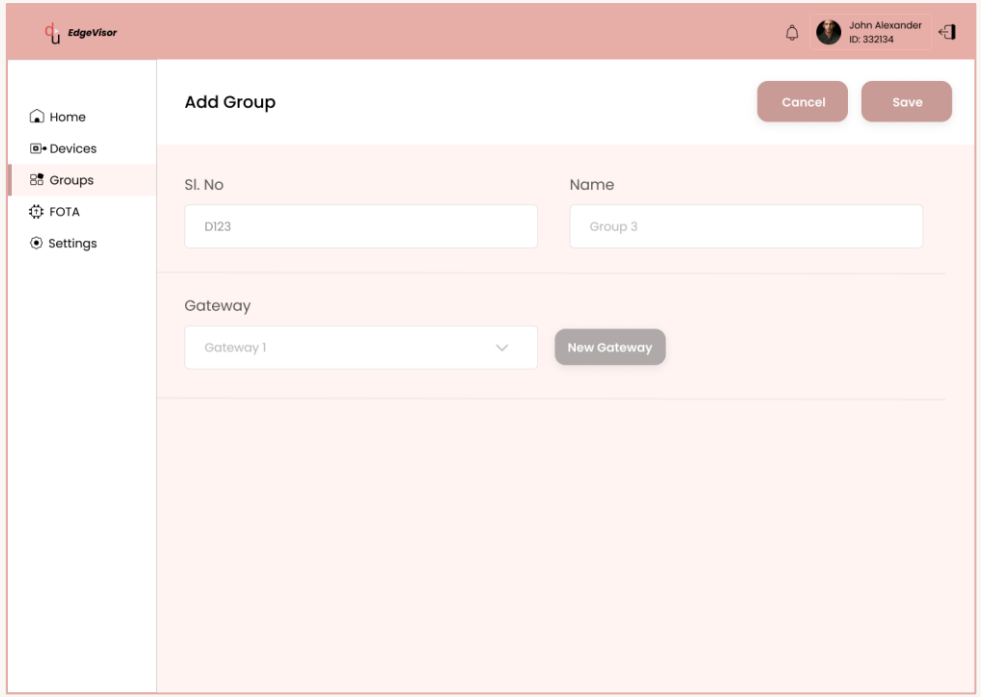
Register at least one M Device and one Gateway.

Resulting state

Newly added devices enter Unconfigured and are ready for grouping or pairing.

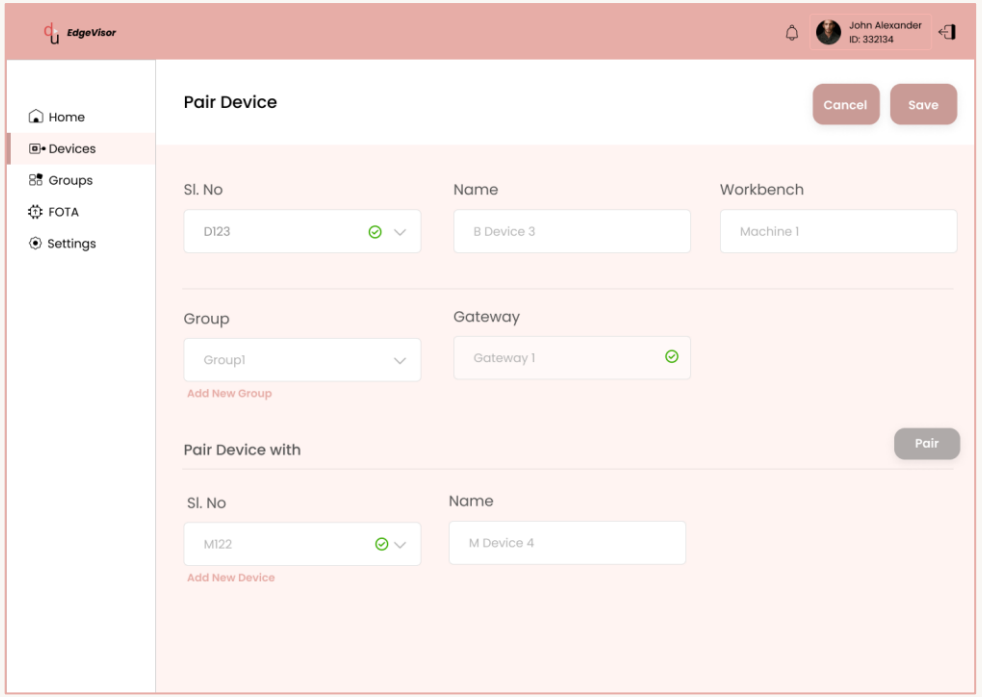
Create the relationship: Gateway group → M Device

Grouping and pairing are dependent steps



06 Create a Gateway group

Open the Gateway action menu, choose Create Group, name it, select the Gateway, and save.

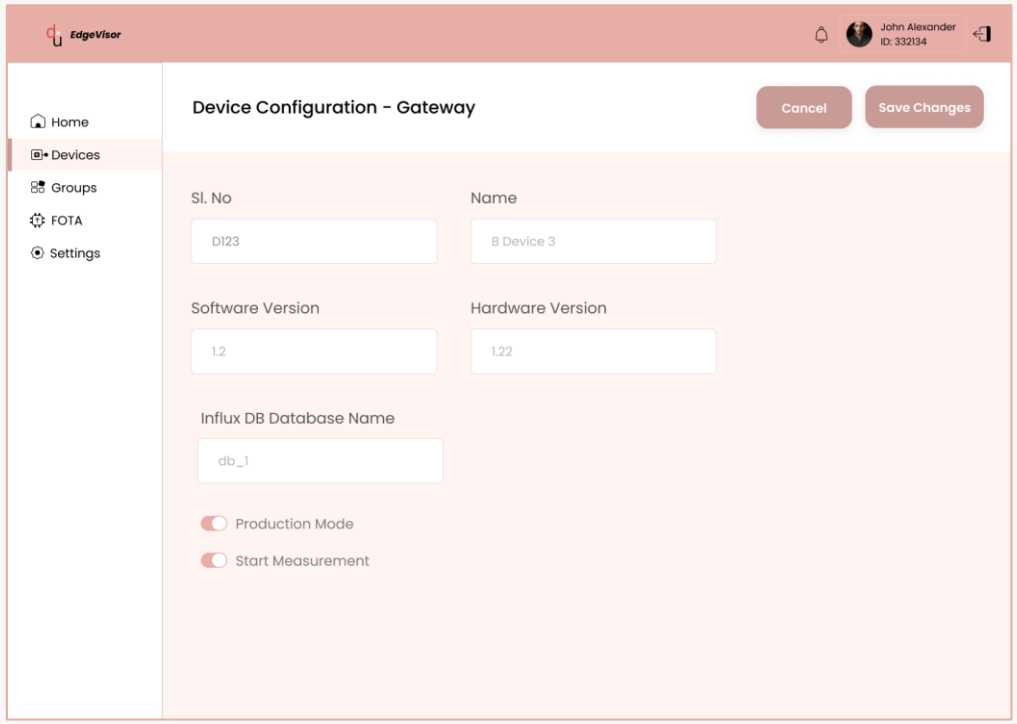


07 Pair the M Device

Open Pair, enter device name and machine, select the group, confirm the linked Gateway, and save.

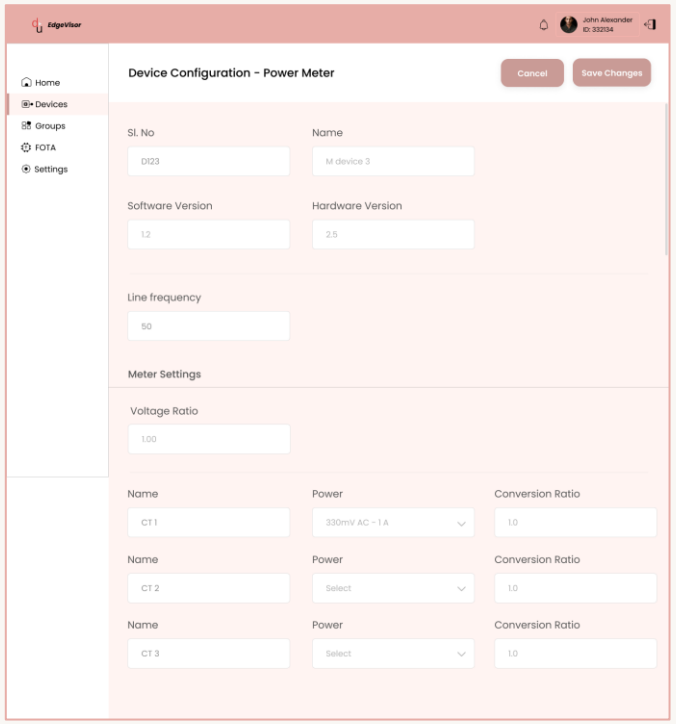
Configure the Gateway and measurement parameters

Both device types must be configured before the workflow reaches Production.



08 Configure Gateway

Enable Start Measurement and save.

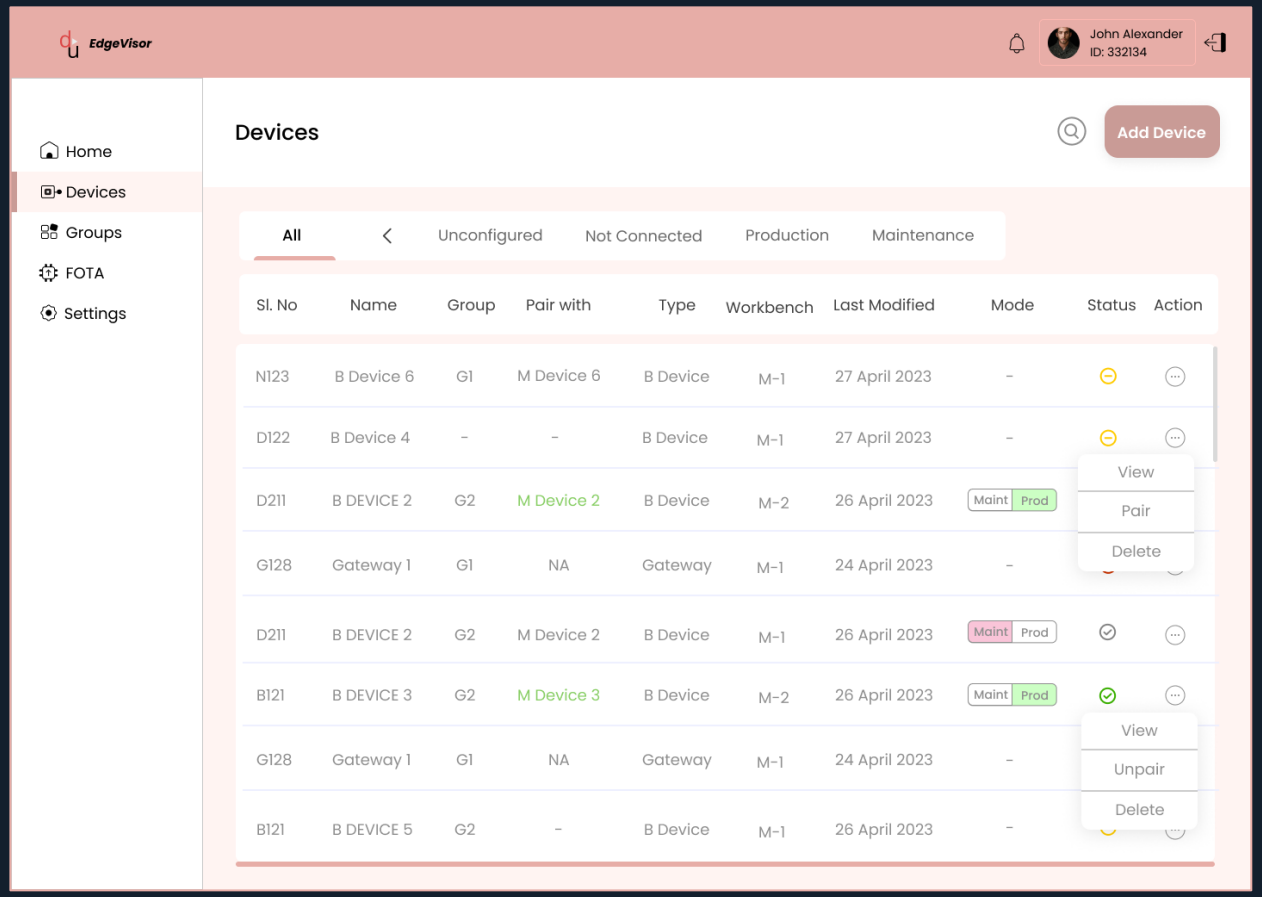


09 Configure M Device

Set Measurement Interval, Line Frequency, and Sensor settings in Meter Settings, then save.

Production is the point where measurement begins

After Gateway and M Device configuration, the M Device captures machine power-consumption data and sends it to the Configurator.



Device state

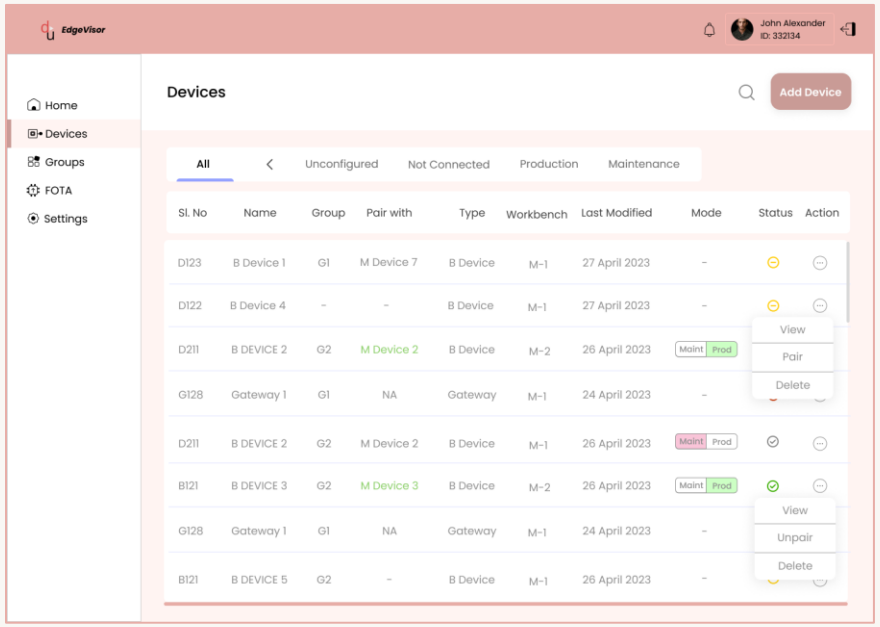
Production

Data path

Machine → M Device → Gateway → Configurator → computer

Maintenance: give operators tools to investigate and recover

To access M Device maintenance, switch the device from Production to Maintenance.



Configure

Adjust M Device settings for measurement.

Unpair

Remove the M Device from its group.

Diagnostics

Check Bluetooth, peripherals, and connectivity.

Log Details

Review previous action logs.

Test

Inspect previously captured power data.

Firmware

Update device firmware.

EdgeVisor provisions devices. DPM Dashboard turns their data into insight.

Once M Devices are in Production, measurement data is available in a separate application for monitoring and analysis.

01

EdgeVisor

Provision and pair M Devices with a Gateway; configure measurement settings.

02

Production data

M Devices capture machine power-consumption data and send it to the Gateway.

03

DPM Dashboard

View Active Power, Energy Consumption, and Productivity across supported views.

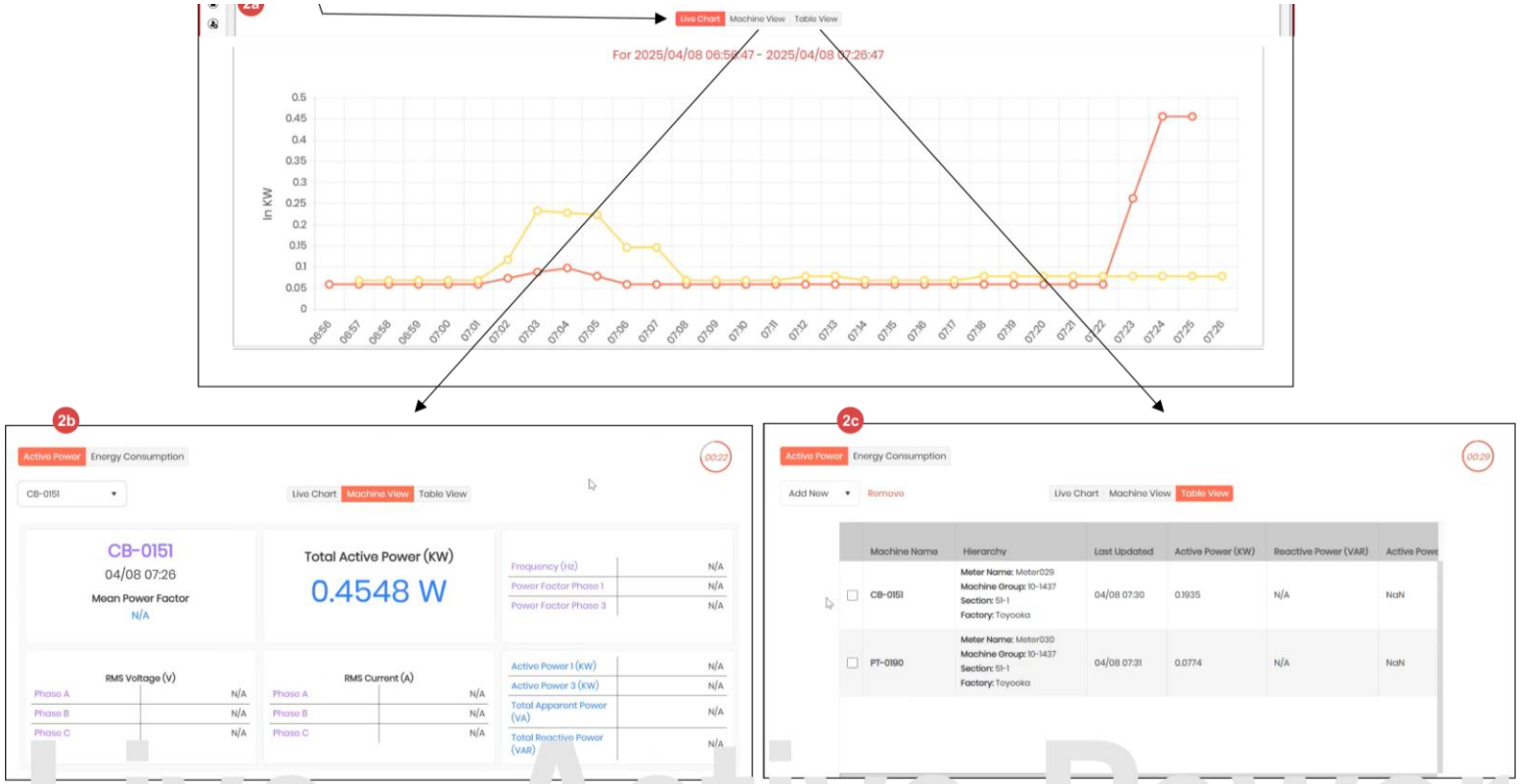


Why this matters

The product experience spans two jobs: preparing connected hardware in EdgeVisor, then reviewing the resulting machine data in DPM Dashboard.

Read live Active Power in three complementary views

The screen shows a time-series chart, a machine-level detail view, and a tabular comparison.



Live - Active Power

Live Chart

Track Active Power over time and inspect changes in the selected time window.

Machine View

Inspect a selected machine's active-power value and related electrical measurements.

Table View

Compare machine records, hierarchy, last-updated time, and power values.

Make power data useful at machine and time-range level

The DPM Dashboard supports distinct metric categories and views for operational monitoring.

Active Power

Current power usage, includes a live chart and machine/table views.

Energy Consumption

Energy usage over a selected period.

Productivity

A separate metric view available from the dashboard tabs.

Time-based analysis described for the dashboard

Second / minute

Hourly

Weekly

Monthly

Example shown in Machine View

Selected machine: CB-0151 • Displayed value: 0.4548 W

Data Captured

The screen labels the metric “Total Active Power (kW)” displays “0.4548 KW.

Designing for connected industry

A central interface that helps operators provision hardware, configure measurement, monitor machine power consumption, and maintain devices.

